Project Planning & Management Document

1. Project Overview

SPIDER – BOT

ADITYA SOMAN AND DHRUV YADAV

This spider bot is a four-legged walking robot designed to mimic basic animal-like movement using servo motors. It is controlled by a microcontroller and moves through coordinated leg motion. The project focuses on understanding how electronic signals can be translated into physical movement.

- *Project Title*
- *Team Members & Roles*
- *Brief Description (2–3 lines)*

2. Team Roles & Responsibilities

Member Name	Role	Responsibilities
ADITYA SOMAN	Initial sketching , prototyping , material procurement, mechanism ideation, product assembly , aesthetic, coding	To make sure the project works smoothly I was responsible for setting up the wiring and ensuring all components like the ESP32, PCA9685, and servos were connected correctly.
DHRUV YADAV	Coding, assembly, initial ideation, video	took charge of setup and testing, adjusting the system until the walking movement felt consistent.

Clearly define who is responsible for what (software, electronics, documentation, etc.)

3. Project Timeline & Milestones

Week	Goals	Deliverables	Status
Week 0	Research reference project, understand components (ESP32, PCA9685, servos), plan basic design prepare 3D model files	Initial concept, component list, basic wiring understanding	Completed
Week 1	Assemble hardware and complete wiring connections Send files for 3D printing	Fully connected circuit with ESP32, driver, and servos Printed parts received and basic structure assembled	Completed

Break project into weekly achievable goals

4. Weekly Progress Log

Week 0

- **Planned Tasks:** Understand project, study reference, finalize components and design
- **Completed Tasks:** Researched working of servos and PCA9685, prepared and finalized 3D design files
- **Challenges Faced:** Understanding how the full system would come together physically
- **Solutions:** Broke the system into smaller parts and focused on one component at a time
- **Next Steps:** Send files for 3D printing and start assembly

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Week 1

- **Planned Tasks:** Get parts printed and begin assembling structure
- **Completed Tasks:** Sent files for 3D printing and assembled the structure once parts were ready
- **Challenges Faced:** Minor fitting/alignment issues with printed parts
- **Solutions:** Adjusted placement and made small manual fixes
- **Next Steps:** Start wiring and electronics setup